

The ratio-closure simplex: literature review and novelty positioning

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Abstract

This document surveys the work that the ratio-closure simplex algorithm has to be positioned against, and states what may and may not be claimed as new. It is deliberately written to *reduce* the claim: every section that identifies a classical or recent antecedent is a section that removes novelty capital, and the positioning in Section 7 is what survives. Claims that earlier drafts made and that this review retracts are listed explicitly in Section 7.1, because a retraction that is not written down tends to reappear.

Every reference was checked against its source; the ones that could not be obtained are listed in `literature/T0\_OBTAIN.md` rather than cited from memory.

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1 Scope and method

1.1 What is being positioned

The object is the ratio-closure simplex algorithm of this project: an ordinary primal simplex method applied to the normalised linear program of the maximum-ratio closure oracle,

$$\max\left\{\sum_i p_i y_i : \sum_i w_i y_i = 1, y_i - y_j \leq 0 \ (i, j) \in \mathcal{A}, y \geq 0\right\},$$

in which a basis is represented by a forest of tight precedence arcs together with one distinguished component, and the outer procedure calls that oracle on successive residual graphs to build the canonical sequence of blocks that yields the LP value of the precedence-constrained knapsack at every capacity.

Four things therefore need positioning, and they have different literatures: the closure model and its LP; the ratio objective; the residual decomposition and its parametric reading; and the simplex-on-a-forest machinery itself. The last is where the strongest antecedents sit, and Section 6 is the section that matters most.

1.2 How the sources were handled

Every reference in this document was checked against its source: journal or publisher page for the classical items, arXiv abstract page for the preprints, the original forum post for the one item that is a forum post. Metadata were not reconstructed from memory. Items that could not be obtained in full text are listed in `literature/T0_OBTAIN.md` with what is needed and why, and are cited here only for facts that the abstract or the record itself supports.

The full texts that were obtained sit in `literature/papers/`. Most come from the project’s own historical library in `PAPER_LONG/BIBLIO`; the recent ones were downloaded from open sources. Each file is named by its BibTeX key — `Yang2026.pdf` for Yang (2026) — so that a citation in this document and the corresponding full text are one lookup apart, and so that the two lists can be reconciled mechanically: `literature/check.sh` reports any PDF without a bibliography entry and any entry without a PDF. `literature/papers/README.md` is the index, with the provenance of each file.

1.3 A note on the direction of this review

A literature review written by the authors of an algorithm has a systematic bias, and the way to counter it is to make the review adversarial on purpose. The organising question of every section below is therefore not *what have others done that is different*, but *what have others already done that we were about to claim*. Section 7.1 lists what that question cost.

2 Maximum closure and its linear program

A *closure* of a directed graph is a set of vertices containing all successors of its members; with the arc convention of this project, $i \rightarrow j$ meaning “ i requires j ”, a closure is a set closed under prerequisites. The maximum-weight closure problem is solved by a minimum cut in a network with infinite-capacity arcs on the precedence relation, a reduction due to Picard (1976) and, in the selection form, to Rhys (1970) and Balinski (1970); Johnson and Niemi (1983) and Gale

(1957) belong to the same early line. The polyhedral fact that makes the reduction work — that the constraint matrix of the closure inequalities is an interval-free network matrix, so the natural LP relaxation is integral — is what allows this project to work with the LP relaxation and still speak about closures.

Picard and Queyranne (1980) characterise the structure of *all* minimum cuts of a network, and Picard and Queyranne (1982) survey the applications of minimum cuts to combinatorial problems, including closure. Both are used here for a specific technical point rather than for background: the distinction between the minimal and the maximal source side of a minimum cut, which is what the canonical convention on tied ratios needs, is theirs.

On the algorithmic side, Hochbaum (2001) revisits minimum cut and maximum flow specifically in closure graphs, Hochbaum and Chen (2000) compare implementations on the open-pit instance of the problem, and Hochbaum (2008) gives the pseudoflow algorithm, which is the method a serious flow-based competitor would use today. Underwood and Tolwinski (1998) is worth singling out: they solve the ultimate pit problem with a network flow algorithm derived from the *dual*, and relate its steps to those of Lerchs–Grossmann, so the idea of attacking maximum closure through dual information is itself old. Amankwah et al. (2014) extend Picard’s reduction to the multi-period case on a time-expanded graph, and Meagher et al. (2014) review the design of pushbacks, which is where the sequence of nested closures produced by the canonical decomposition is actually consumed. Ahuja et al. (1993) and Schrijver (2003) are the standard references for the surrounding network flow and combinatorial optimisation material.

A generalisation worth recording, because it places the closure problem inside a larger family rather than beside it: Hochbaum and Queyranne (2003) minimise a convex cost over a closed set, which specialises to maximum closure on one side and to isotonic regression on the other. The same family is visited from the other end by the maximum density subgraph, which is the ratio objective of Section 3 without any precedence structure at all — solved by flow in Goldberg (1984), given its linear program by Charikar (2000), and treated as one parametric cut by Hochbaum (2024).

Not everything in this line is a flow algorithm, and the exception matters enough to have its own place: the algorithm of Lerchs and Grossmann (1965) maintains a *forest* and pivots on it. It is discussed in Section 6.1, next to the recent forest-structured work, because that is where it does its damage.

What this costs the claim. Everything about the closure model, its LP relaxation, its integrality and its solution by minimum cut is classical. Nothing in ratio-closure simplex may be presented as new at this level, and the project’s own documents say so: the mathematical contract states the model and attributes it, and the tutorial recalls the reduction only to fix notation.

3 The ratio objective

Two classical devices remove a ratio from an objective. Charnes and Cooper (1962) rescale a linear fractional program into a linear one by normalising the denominator, which is exactly the transformation that turns $\max p(S)/w(S)$ into the program with $\sum_i w_i y_i = 1$ that ratio-closure simplex solves. Dinkelbach (1967) instead treats the ratio parametrically, solving $\max\{p(S) - \lambda w(S)\}$ and updating λ by Newton’s method; applied to closures, each iterate is a maximum-closure problem and therefore a minimum cut. This project implements the second as one of its reference backends.

Maximum-ratio closure — maximum-density or maximum-mean closed set, depending on the community — is therefore classical in both readings, and the equivalence between the fractional point of view and the level-set argument ($y = \int \chi^{C_t} dt$, each C_t a closure) is the standard device

relating fractional closure points to closures. The tutorial of this project proves it in that form and attributes it there.

3.1 The LP class, stated precisely

Since the positioning turns on which linear program is being specialised to, it is worth naming the class rather than the application. The program is

$$\max \sum_{i \in R} p_i y_i \quad \text{s.t.} \quad \sum_{i \in R} w_i y_i = 1, \quad y_i - y_j \leq 0 \quad \forall (i, j) \in A[R], \quad \mathbf{y} \geq 0, \quad (1)$$

with $\mathbf{w} > 0$, $\mathbf{p}$ of arbitrary sign and no upper bounds. Three equivalent readings of the same class:

- (a) **One budget row over a homogeneous difference system.** The constraints $y_i - y_j \leq 0$ form a system of homogeneous difference inequalities on a DAG; a single dense equality row with positive coefficients is added.
- (b) **The base of an order cone.** The set $\{\mathbf{y} \geq 0 : y_i \leq y_j\}$ is the cone of nonnegative order-preserving vectors on the poset, whose extreme rays are the characteristic vectors of the nonempty closed sets; the equality slices it into the polytope whose vertices are $\chi^C/w(C)$. Optimising over it therefore returns $\max_C p(C)/w(C)$, which is why the transformation of [Charnes and Cooper \(1962\)](#) is exact here. The polytope is the cone analogue of Stanley’s order polytope, whose vertices are the characteristic vectors of the filters of the poset ([Stanley, 1986](#)).
- (c) **Transposed incidence plus one dense row.** Each difference constraint has a $+1$ and a -1 in the columns of two vertex variables, so the matrix is the *transpose* of a digraph incidence matrix, with one dense row appended and a right-hand side that is zero on the network part.

Reading (c) matters for Section 6 and was stated loosely in an earlier draft. In the side-constrained network flow literature the variables are arcs and the rows are vertices; here the variables are vertices and the rows are arcs. The two are linear programming duals of one another — the dual of (1) is $\min \beta$ subject to $w_i \beta + \text{div}_i(\boldsymbol{\alpha}) \geq p_i$, $\boldsymbol{\alpha} \geq 0$, a flow-like system with one scalar — so the algebra transports between them, but pricing on one side corresponds to the ratio test on the other and the roles are not interchangeable without care. This is the same duality [Underwood and Tolwinski \(1998\)](#) exploit when they attack the ultimate pit problem through the dual. It also places (1) in the same structural family as isotonic regression and total variation on a graph, which are difference systems on vertex variables too — and that is why [Yang \(2026\)](#) is closer prior art than the network-simplex-with-a-side-row line, not further.

The one non-generic modelling choice. Replacing the equality $\sum_i w_i y_i = 1$ by an inequality adds the origin to the feasible set and returns value zero whenever every nonempty closure has negative profit. With nonnegative profits this is immaterial. Here it is not: profits carry either sign, a profitable vertex may require loss-making prerequisites, and the outer decomposition calls the oracle on residual graphs whose profits deteriorate, so the last calls typically have a negative optimal ratio and the decomposition still needs the *best nonempty* closure at each of them. The equality is deliberate and is argued as such — but it is a modelling choice inside a classical transformation, not a contribution.

Recent work on the ratio side. [Hochbaum \(2024\)](#) gives an incremental parametric cut method for ratio problems of this family; it is the most recent point of comparison found for the oracle itself and is in `literature/papers/`. It is a competitor to the Dinkelbach backend, not to Ratio-closure simplex’s basis machinery, and it should be cited where the oracle is discussed.

4 The residual decomposition and its parametric reading

Repeatedly extracting a maximum-ratio set and continuing on the residual is not a device of this project. It is the decomposition of [Sidney \(1975\)](#) for precedence-constrained single-machine scheduling, developed in the sequencing literature by [Lawler \(1978\)](#) and surveyed by [Correa and Schulz \(2005\)](#); [Margot et al. \(2003\)](#) give the version closest to the ratio reading used here: an in-depth study of an LP relaxation of a precedence-constrained problem whose algorithmic core is a parametric extension of Sidney’s decomposition, obtained by essentially solving a parametric minimum-cut problem.

The vocabulary of this document follows [Dose et al. \(2026\)](#), the companion work of this project, which studies the same canonical object on a precedence graph that is a directed forest. What is called a *canonical block* here is a *closure layer* there; the canonical decomposition is their *optimal sequence of closure layers*; the ratio λ_r of a block is a *breakpoint* of $u(\lambda)$; and the per-vertex bound this project computes is their *threshold* of a vertex — the same quantity, not an analogy. Their scalar λ , parametrising the weights as $p_i - \lambda w_i$, is the ρ of the ratio oracle here, which is also the reduced cost $b_i = p_i - \rho w_i$ of Section 6.1. The two works do not overlap on content: they give aggregation algorithms specialised to forests, reaching $\mathcal{O}(n \log n)$, whereas the precedence graph here is an arbitrary DAG and the method is the simplex. The sequence produced is the same object, so the two are directly comparable on instances that happen to be forests — and on those instances their bound is the one to beat. The same object appears from the flow side as the sequence of nested minimum cuts of a parametric network: [Eisner and Severance \(1976\)](#) give the divide-and-conquer scheme that computes all breakpoints with a number of maximum-flow computations proportional to their number, and [Gallo et al. \(1989\)](#) give the parametric maximum-flow algorithm that computes the whole path in the time of a single flow computation for monotone parametrisations. [Topkis \(1978\)](#) supplies the lattice/submodularity view that makes the nesting structural rather than incidental. [Picard and Smith \(2004\)](#) apply exactly this to open-pit contours, and [Eppstein \(2018\)](#) gives the modern complexity treatment of the parametric closure problem.

This project implements the Eisner–Severance scheme as its **parametric** backend, and its experimental report is explicit that the implementation is a divide-and-conquer decomposition with independent maximum flows, without the amortised flow reuse of [Gallo et al. \(1989\)](#).

What this costs the claim. The existence of the canonical decomposition, its nesting, the strictly decreasing ratios, the interpolation between two consecutive prefixes that yields the LP value at a given capacity, and the solution of the whole path by parametric flow, are all established. The project’s mathematical contract already presents them as such. A claim of novelty at this level would be simply wrong, and the earlier drafts were right to drop it.

[Nemesch et al. \(2025\)](#) survey exact and approximation algorithms for linear-parametric optimisation and place the closure case inside the general picture; it is the reference to consult before asserting that any parametric technique used here is unusual.

What survives here. Only one thing, and it is small: the convention that resolves ties. When several closures attain the maximum ratio, the canonical block is their union, which on the flow side is the *maximal* source side of a minimum cut rather than the minimal one ([Picard and Queyranne, 1980, 1982](#)). This project implements both conventions and verifies that they agree; that is careful engineering on a known structure, not a new result.

5 The precedence-constrained knapsack and open-pit mining

The problem the outer procedure solves is the LP relaxation of the precedence-constrained knapsack, which has two largely separate literatures.

Polyhedral and exact work. The PCKP polytope has been studied through induced cover and clique inequalities and their lifting (Boyd, 1993; Park and Park, 1997; van de Leensel et al., 1999), with separation of maximally violated inequalities (Cho and Shaw, 1997; Shaw and Cho, 1998) and exact algorithms (Samphaiboon and Yamada, 2000); Kellerer et al. (2004) is the standard monograph, and Kolliopoulos and Steiner (2007) treat the approximation side. Tree-structured special cases have their own line (Johnson, 1968; Horn, 1972; Adolphson and Hu, 1973). This literature is about the integer problem; ratio-closure simplex solves the LP relaxation, so it borders on this work rather than competing with it. The distinction matters and the project’s documents keep it: the solver treats the LP, the integer problem is a different problem.

Mining. The ultimate-pit problem is maximum closure (Lerchs and Grossmann, 1965; Matheron, 1975; Vallet, 1976), and the capacitated and multi-period versions are where the PCKP LP relaxation is actually used at scale. Bienstock and Zuckerberg (2010) give the decomposition method for LP relaxations of large precedence-constrained problems that is the reference point for scale, studied further by Muñoz et al. (2018); Chicoisne et al. (2012) give an algorithm for the open-pit production scheduling problem; Boland et al. (2012) and Espinoza et al. (2015) are the surrounding work, the latter being the MineLib benchmark this project derives instances from. Byun and Dimitrakopoulos (2013) solve the single-capacity closure LP by parametric flow, which is the closest published match to the task Ratio-closure simplex performs.

What this costs the claim. Two things. First, the single-capacity closure LP by parametric flow is already published, so ratio-closure simplex enters a solved problem and must justify itself on how it solves it, not on what it solves. Second, at the scale where this LP matters in practice, the incumbent methods are flow-based and mature. The project’s own experiments are consistent with that: on the instances derived from MineLib the forest configurations solve nothing within the time limit while the parametric backend solves all of them.

6 Algorithms whose state is a forest

This is the section that decides the claim. Four bodies of work sit between Ratio-closure simplex and any statement of the form “an algorithm whose state is a forest on the precedence graph”. The first of them is sixty years old and comes from mining, which is where this project’s own instances come from; it was missing from the earlier drafts of this review, and it is the most uncomfortable of the four.

6.1 Normalized trees: Lerchs–Grossmann, 1965

Lerchs and Grossmann (1965) solve maximum closure with a graph algorithm, not with a flow algorithm, and the object it maintains is a forest. In the exposition of Hochbaum (2001), from which the following is taken verbatim in substance: the algorithm maintains a spanning tree rooted at an added vertex r in the extended graph, called a *normalized tree*; each branch T_q hanging from the root carries its *mass* $M_q = b(T_q)$, the sum of the vertex weights it contains; a downward arc is *strong* if the mass it supports is positive; the tree is normalized when its only strong arcs are the ones adjacent to r ; and, dropping the root arcs, Hochbaum writes that “we also refer to the forest (V, A_T) in G as a normalized tree”. The invariant is that the union of the strong branches is a maximum closed set of the forest, and the weight of the strong vertices bounds the maximum closure value from above.

An iteration is a pivot in everything but the name. Quoting Hochbaum (2001) again: a *merger* consists of “adding an arc (s, w) from a strong branch S to a weak branch W and removing the arc from the root of the branch S to the root r ”, after which the branch hangs from the strong

vertex s and the supported masses are updated. One arc enters, one arc leaves, one component loses its root, and the aggregates on the affected path are recomputed. Zhao and Kim (1992) is a variant that differs in how the blocks are regrouped after a violation, which is the split side of the same operation, and Giannini et al. (1991) is the flow-based implementation that competed with it.

Two further facts from Hochbaum (2001) matter here. First, the normalized tree carries enough information to construct a feasible flow in linear time, so the primal–dual correspondence between the forest state and a flow certificate is known. Second, he gives a parametric version of the Lerchs–Grossmann algorithm with the same complexity as a single run — which makes it a *competitor* on the task ratio-closure simplex performs, not merely an antecedent. Picard and Smith (2004) close the loop on the objective as well: they compute intermediate open-pit contours maximising a benefit–cost ratio by parametric maximum flow.

What this costs the claim. A great deal, and more than the recent work does. The correspondence with ratio-closure simplex is not thematic but term by term: a branch aggregate $b(T_q)$ against the reduced cost $p(T) - \rho w(T)$; a strong arc against a positive reduced cost; the union of the strong branches against the support of the current basic solution; a merger against an entering arc with its leaving root arc. Setting $b_i = p_i - \rho w_i$ makes the Lerchs–Grossmann strong/weak test literally the sign of the ratio-closure simplex reduced cost. So the forest state, the branch aggregates, the sign test on an aggregate and the merge operation are not new relative to the 2025–2026 work: they are not new relative to 1965. What Lerchs–Grossmann do *not* do is run a simplex — there is no basis in the LP sense, no ratio test, no dual multiplier as an LP object — and they solve the closure problem at fixed weights rather than the ratio LP. That distinction is now the whole of the surviving claim, and Section 8 Q5 says what has to be proved to keep it.

6.2 Network simplex with a side constraint

That a basis of a network LP is a spanning tree, that adding a side row makes it a spanning structure plus one extra element, and that pivots are described as operations on that structure, is textbook material (Ahuja et al., 1993; Schrijver, 2003). The paradigm “simplex on an incidence matrix with one side row” is therefore not available as a contribution, and any formulation that suggests otherwise must be rewritten.

The specific case of one side row has its own line. Chen and Saigal (1977) give a revised simplex variant for capacitated network flow with additional linear constraints in which the working basis has the size of the side system rather than of the network. Çalışkan (2011) specialises the network simplex to the constrained maximum flow problem. Holzhauser et al. (2017) give a fully combinatorial network simplex for the budget-constrained minimum cost flow problem, with two kinds of vertex potentials, three kinds of reduced costs, optimality criteria proved directly on the network and an explicit anti-cycling treatment. These are the right block to read, but they are one transposition away from (1): their variables are arcs and their rows are vertices, whereas here the variables are vertices and the rows are arcs (Section 3.1(c)). The two formulations are linear programming duals, so the machinery transports and the differences have to be argued rather than assumed (Section 8, Q3) — but the resemblance is a duality, not an identity, and an earlier draft overstated it.

6.3 Yang: a specialised simplex on rooted spanning forests

Yang (2026) gives, for budget-constrained total-variation-regularised linear programs on graphs, a characterisation of basic solutions in terms of rooted spanning forests with orientation, an interpretation of the simplex method as operations on those forests, and an accelerated implementation compared against a commercial solver. The abstract states the first two points directly.

Total variation on a graph is a difference system on vertex variables with a budget row, so his program and (1) are in the same structural family, and not merely in the same spirit: this is the closest match in formulation type among all the work reviewed here.

The overlap with ratio-closure simplex is substantial and was underestimated in an earlier draft. Yang’s setting has a budget row and a graph-structured constraint matrix; when the budget slack is nonbasic he obtains a component without a root, which plays the role of the rootless positive component here; he writes the basic variables explicitly in terms of the nonbasic ones, derives reduced costs for root and edge variables, states optimality in terms of tree and subtree shifts, reads pivots as merges, splits and shifts, uses blocking edges, and updates subtree aggregates locally.

Consequently the following claim, made in an earlier draft, is withdrawn: that no work carries revised-simplex algebra to this level of explicitness on a forest structure. Yang does exactly that, on a different linear program.

6.4 Wuille and the spanning-forest linearization line

Wuille (2025) sets up, for Bitcoin transaction cluster linearization, the program $\max \sum_i f_i t_i$ subject to $\sum_i s_i t_i = 1$, the dependency inequalities and $t \geq 0$: the same normalised program ratio-closure simplex solves, up to orientation convention and application. The post states that the method maintains $n - 1$ free variables, that the active dependencies form a spanning forest, that within every chunk there is at most one free vertex variable, that all chunks except one contain such a variable and are therefore excluded, and it describes the pivot operations as merging and splitting chunks. These were verified against the post itself.

Mollakhani et al. (2026) formalise that reading: the paper presents the equivalent linear programming formulation and develops the Spanning Forest Linearization algorithm, motivated, in the authors’ words, by structural properties of basic feasible solutions of the simplex method represented through spanning forests. It then leaves the exact simplex: it stops maintaining the identity of the root and a single distinguished included component, compares chunks locally, and optimises the linearization directly.

This is the closest antecedent in absolute terms, and it removes several items from the candidate contribution list: the normalised program itself, the count of $n - 1$ nonbasic variables, the forest of tight slacks, the “at most one free vertex variable per component”, the single positive component, the connectivity of the support, and the four qualitative pivot effects. None of these may be presented as new.

6.5 What the four leave open

Lerchs–Grossmann maintain a forest on the closure graph and test the sign of branch aggregates, but do not run a simplex and do not solve the ratio LP. Yang (2026) develops the exact simplex, but for a different program. Wuille and Mollakhani et al. work on the same program, but leave the exact simplex in favour of a direct linearization algorithm. Holzhauser et al. (2017) develops a combinatorial simplex for a network with one side row, but on minimum cost flow. The intersection of all four — the ordinary primal simplex, developed exactly, on the maximum-ratio closure program — is where whatever remains of the contribution has to live, and Section 7 states how little that is and what would have to be shown to make it stand.

7 Positioning: what may be claimed

7.1 Retracted claims

The following statements appeared in earlier drafts of this project and are withdrawn. They are recorded here rather than silently deleted, because a retraction that is not written down tends to

reappear in the next draft.

- R1. *“First complete closed-form realization of the two revised-simplex basis systems for a forest-structured graph LP.”* Withdrawn: too close to Yang (2026), who develops exactly that on a budget-constrained total-variation program.
- R2. *“First closed-form forest simplex.”* Withdrawn: too broad, and contradicted both by Yang (2026) and by the classical network-simplex-with-side-constraint literature (Ahuja et al., 1993; Schrijver, 2003).
- R3. *Novelty of the forest structure of basic feasible solutions.* Withdrawn: present in Wuille (2025) and formalised in Mollakhani et al. (2026) for the same normalised program.
- R4. *Novelty of the merge/split reading of pivots.* Withdrawn: same sources, and Yang (2026) independently.
- R5. *Novelty of the forest state and of the aggregate sign test, even relative to the 2025–2026 work.* Withdrawn, and further back than the previous two items: Lerchs and Grossmann (1965) maintain a forest of branches on the closure graph, carry the mass of each branch, classify an arc by the sign of the mass it supports, and pivot by adding one arc and removing the branch’s root arc (Section 6.1). With $b_i = p_i - \rho w_i$ their strong/weak test is the sign of the ratio-closure simplex reduced cost. Nothing about maintaining a forest and testing aggregates on it can be claimed.
- R6. *Novelty of the maximum-ratio closure formulation, of its solution by max-flow, and of the canonical decomposition.* Withdrawn: classical (Picard, 1976; Picard and Queyranne, 1982; Sidney, 1975; Lawler, 1978).
- R7. *Novelty of the “simplex on an incidence matrix plus one side row” paradigm.* Withdrawn, and datable. Chen and Saigal (1977) prove that a basis of a capacitated flow problem with k additional linear rows is a spanning tree plus k columns, and drive pricing, pivoting and inversion from a working basis of size k ; Çalıřkan (2011) specialises the case $k = 1$ to maximum flow with a budget row, where the basis is a spanning tree plus one arc closing a fundamental cycle of nonzero net cost; Holzhauser et al. (2017) do the same for minimum cost flow. The paradigm is due to 1977, not to the recent literature.
- R8. *Novelty of solving the ratio version by walking a forest state.* Withdrawn: Hochbaum (2001) gives a parametric Lerchs–Grossmann with the complexity of a single run, and Picard and Smith (2004) compute open-pit contours maximising a benefit–cost ratio by parametric flow.
- R9. *Novelty of the equality normalisation $\sum_i w_i y_i = 1$.* Withdrawn: it is the homogeneous case of the reduction of Charnes and Cooper (1962), whose Theorem 2 already writes the normalisation as an equality with right-hand side 1 (Section 8, Q5).
- R10. *Any claim of superior practical performance.* Withdrawn on the project’s own evidence: on the frozen test bed the parametric and Dinkelbach max-flow backends dominate ratio-closure simplex on both solved count and time (Section 8).

7.2 The cell nobody occupies

The claim reduces to one empty cell in a table with two axes: what the method maintains, and which program it solves.

	maximum closure, weights fixed	maximum-ratio closure
flow / cut state	Picard (1976) , Hochbaum (2008)	Gallo et al. (1989) , Hochbaum (2024)
forest state, pivots, no LP basis	Lerchs and Grossmann (1965)	Hochbaum (2001) (parametric LG), Mollakhani et al. (2026)
exact simplex, LP basis	—	this work
<i>exact simplex on a sibling program</i> : Yang (2026) (budget-constrained total variation), Holzhauser et al. (2017) , Chen and Saigal (1977) (k side rows on a capacitated flow), Çalışkan (2011) (budget-constrained max flow)		

Read across the bottom row: the exact simplex has been developed on programs in the same structural family as (1) — a difference system or a network with one budget row — but not on (1) itself. Read up the right column: the ratio program has been solved by flow states and by forest states that pivot, but not by a simplex maintaining an LP basis with reduced costs, a ratio test and dual multipliers. [Mollakhani et al. \(2026\)](#) say so themselves: their algorithm is, in the abstract’s own words, motivated by structural properties of basic feasible solutions of the simplex method, and what it then maintains and refines is a partition of the vertices into dependency-respecting subsets ordered by decreasing aggregate ratio — chunks merged and split directly on the graph, not a basis.

The most serious candidate for the empty cell, and why it does not fill it. [Margot et al. \(2003\)](#) is the one paper found in this review that promised to occupy the bottom-right cell: an in-depth theoretical, algorithmic and computational study of an LP relaxation of a precedence-constrained problem, whose algorithmic core is a parametric extension of Sidney’s decomposition and whose finest decomposition is obtained, in their words, by essentially solving a parametric minimum-cut problem. Read in full, it belongs in the first row and not in the third: the state it maintains is a sequence of nested minimum cuts, and the only simplex in the paper is the one inside the general-purpose LP solver used to produce the computational results. It carries no basis characterisation, no reduced costs and no ratio test for the relaxation. It strengthens R6 and R8 — the decomposition and its parametric reading are classical — and leaves the cell empty.

A warning that belongs next to the table. An empty cell is not by itself a contribution. It can be empty because the route is worse, and the evidence available here points that way: on this project’s own test bed the flow-based entries of the first row and the parametric entry of the second dominate the bottom row on both solved count and time (Section 8). The honest reading is that this work fills the cell and reports what is inside it, including the part that is unfavourable.

7.3 What is left

Four candidates survive the review. None of them is a paradigm; all of them are specialisations, and each carries a condition that has not yet been discharged.

C1. Specialisation of the ordinary simplex, primal and dual, to the normalised maximum-ratio closure LP, and to the canonical decomposition it drives. That a simplex method is exact is not a contribution and is not offered as one; what is at stake is what the method *becomes* on this program. [Wuille \(2025\)](#) and [Mollakhani et al. \(2026\)](#) expose the forest structure of the basic feasible solutions of this program and then leave the simplex, optimising the linearization directly; [Yang \(2026\)](#) develops the exact simplex, but on a different program; [Lerchs and Grossmann \(1965\)](#) pivot on a forest of the closure graph, but outside the simplex and at fixed weights. Carrying the ordinary simplex through on this LP — basis state, prices, pivot directions, ratio test, multipliers, certificates, in a

primal and in a dual variant — and then driving the whole canonical sequence of blocks with it, not merely the first, is the gap between all three. *Condition:* it must be presented as a specialisation, not as a new method; the debt to all three lines must be stated in the abstract, not only in related work; and the difference from a normalized tree must be made precise rather than asserted, because to a mining audience the two objects look identical.

- C2. **The closure-specific closed forms are markedly simpler than the general forest-simplex algebra.** For the normalised program the reduced costs collapse to

$$\bar{c}_{y_T} = p(T) - \rho w(T), \quad \bar{c}_{s_e} = \sigma_e [p(D_e) - \rho w(D_e)],$$

with no edge objective coefficients, no box-bound contributions, no orientation choice between positive and negative edge variables and no generic working-basis algebra. *Condition:* the comparison against Yang (2026) must be made explicitly and term by term; “simpler” is not a claim until the two sets of formulas are put side by side.

- C3. **Full pricing in $O(n)$ arithmetic, independent of the number of precedence arcs.**

There are exactly $n - 1$ nonbasic candidates; all forest aggregates are obtained in one postorder; each candidate costs $O(1)$; no non-forest arc is ever priced. *Condition:* this is the cleanest of the four, and also the one most exposed. If Yang (2026) admits an equivalent $O(n)$ pricing in his model, the claim is not that $O(n)$ pricing is possible but that it is available here without the structural overhead his model carries. That check has not been done.

- C4. **The separation between pricing and feasibility in the exact ratio test.** Pricing ranges over the $n - 1$ nonbasic forest and root variables; feasibility ranges over all potentially decreasing basic slacks. The distinction is clean and it is what makes the exact simplex correct rather than merely plausible. *Condition:* on its own this is a component of C1, not an independent contribution, and should be presented as such.

7.4 The formulation that survives

To the best of our knowledge we give the first explicit simplex realisation, primal and dual, specialised to the normalised maximum-ratio closure LP, and we use it to produce the entire canonical sequence of blocks rather than the maximum-ratio block alone. Building on the forest structure of its basic feasible solutions — which is closely related to recent spanning-forest linearization work — we derive problem-specific closed forms for all nonbasic prices and pivot directions. In particular, complete pricing requires $O(n)$ arithmetic independently of the number of precedence arcs, while a full ratio test accounts for all potentially blocking basic slacks. The canonical decomposition itself is classical and is not claimed.

followed immediately, and in the same place, by:

We do not claim novelty for the forest structure itself, for the merge/split basis effects, for the maximum-ratio closure formulation, for the canonical decomposition, or for the general paradigm of simplex methods on incidence matrices with side constraints; these have direct precedents in Wuille (2025) and Mollakhani et al. (2026), in Yang (2026), and in the classical network-simplex literature.

7.5 The name

The name follows the positioning, and the earlier one contradicted it. “Forest Simplex” named the forest — the one element Section 7.1 withdraws — and it collided three ways: with the

spanning forest of Mollakhani et al. (2026) and Wuille (2025), with the *rooted spanning forests* of Yang (2026), and, one letter and no syllables away, with the Forrest–Tomlin update of the basis factors, which is what an audience of linear programmers hears first.

The method is therefore called the ratio-closure simplex, lower case, on the model of *network simplex*: a name taken from the problem the method is specialised to, not from the structure it maintains. The problem is $\max_C p(C)/w(C)$ over the closures of a digraph, and the linear program is (1); the capacity row of the application is absent from both, which is why it is absent from the name as well. The precedence-constrained knapsack is where the program comes from and where the method is measured, and it is reported as such among the applications; it is kept out of the title as well as out of the name, because the method delivers the whole canonical sequence and not only the first block, and a knapsack in the title would promise that the capacity row is handled by the simplex, which it is not. The forest keeps its own name, as the structure of the basis, with the attribution Section 6.1 gives it.

7.6 The question the work answers

The question is not whether maximum-ratio closure can be solved with forests — it can, and that is known. It is:

what does the exact ordinary primal simplex look like on the normalised maximum-ratio closure LP, once the known forest structure of its basic feasible solutions is exploited completely?

The answer is the algorithm: basis state as forest plus root, exact aggregate prices, $O(n)$ pricing, exact forest motion, full ratio test, dual multipliers and certificates, and an implementation measured against the alternatives. That last item is where the work currently stands weakest, and Section 8 says why.

8 What has to be settled before claiming

8.1 The checks this review could not close

- Q1. **Term-by-term comparison with Yang (2026).** Contribution C2 asserts that the closure-specific closed forms are markedly simpler than the general forest-simplex algebra, and C3 asserts that $O(n)$ pricing is available here without the structural overhead of Yang’s model. Neither has been verified against the paper’s actual formulas. This is the single most load-bearing open item: if Yang’s development specialises to something equivalent, C2 and C3 both collapse into “an instance of a known scheme”.
- Q2. **Exact delimitation against Mollakhani et al. (2026).** Their abstract settles the direction: the algorithm is motivated by the structure of basic feasible solutions of the simplex method, and it then merges and splits chunks directly on the dependency graph, maintaining a partition ordered by decreasing aggregate ratio rather than an LP basis. That is a departure from the simplex, not a specialisation of it. What remains unchecked is the full text: whether the paper anywhere derives reduced costs, a ratio test or dual multipliers for (1), and how far its optimality argument coincides with the pricing argument used here.
- Q3. **The network simplex with one side row.** *Closed on the structural question; open only on the term-by-term comparison.* Chen and Saigal (1977) and Çalıřkan (2011) have now been obtained and read. They push this block back to 1977 and settle what it does and does not cover.

Chen and Saigal (1977) treat a capacitated flow problem with k arbitrary additional linear rows. Their Proposition 1.1.5 is the structural statement: in any partition of a basis, the

square block corresponds naturally to a *spanning tree* of the network, so a basis is a spanning tree plus k further columns, and the subnetwork carried by the basis has rank $n - 1$ and nullity k . Pivoting, pricing and inversion are then driven by a *working basis* of size k rather than $n + k - 1$. Çalışkan (2011) is the case $k = 1$ made explicit on maximum flow with a budget row: a basis is the two subtrees rooted at s and t , the arc (t, s) , and *one* further arc closing a fundamental cycle, with his Lemma 1 recording that the net cost around that cycle cannot vanish, which is the rank condition the side row imposes.

Two consequences. First, R7 in Section 7.1 is confirmed and can now be cited precisely instead of being called textbook: tree-plus- k bases, a working basis of the size of the side rows, and a specialised pricing built on them are due to 1977, not to 2017. Second, the phrase “to the best of our knowledge” in Section 7 survives, for a reason Section 3 already states: this whole block works on the node–arc incidence matrix with variables on *arcs*, where a basis is a spanning tree, while (1) carries variables on *vertices* and difference constraints on arcs — the *transposed* incidence matrix — where a basis is a forest with anchored roots and not a spanning tree of the same graph. The paradigm is theirs; the specialisation to the transposed structure is not in any of the three papers.

What remains open is narrower and is the same check as Q1: Holzhauser et al. (2017), the closest of the block, has been obtained but not read term by term, so the claim that the closure-specific closed forms are simpler than the generic working-basis algebra is still asserted rather than demonstrated for that paper.

- Q4. **Formal separation from the normalized tree.** This is now as load-bearing as Q1. A normalized tree of Lerchs and Grossmann (1965) and an ratio-closure simplex basis are, on inspection, the same object read twice: forest plus root, aggregates per branch, a sign test per branch, one arc in and one root arc out. What has to be written down is the exact statement of the correspondence and of where it breaks — presumably that a normalized tree is not a basis of the ratio LP, that it carries no ratio test and no dual multiplier, and that the parametric Lerchs–Grossmann of Hochbaum (2001) walks breakpoints rather than performing simplex pivots on the normalised program. If instead the correspondence turns out to be exact, then ratio-closure simplex is a reinterpretation of a 1965 algorithm, which is a publishable observation but a completely different paper from the one currently drafted.
- Q5. **Prior art on the equality normalisation.** *Closed, and it is a precedent.* Charnes and Cooper (1962) has now been obtained and read. Their Theorem 2 reduces a linear fractional program to the ordinary linear program

$$\max c^T y + \alpha t \quad \text{s.t.} \quad Ay - bt \leq 0, \quad d^T y + \beta t = 1, \quad y, t \geq 0,$$

in which the normalisation is an *equality* and the constant is exactly 1. The program (1) that ratio-closure simplex solves is the homogeneous case $\alpha = \beta = 0$, $d = w$, of that transformation. The deliberate choice of an equality is therefore not a modelling contribution of this project; it is the 1962 construction, and Section 3 already cited it as such.

Their treatment of sign is worth recording because it is *not* ours. Charnes and Cooper require $\text{sgn}(\gamma) = \text{sgn}(d^T x^* + \beta)$ and therefore solve two linear programs, the second obtained by negating (c, α) and (d, β) — “a change in sign in the functional and in one constraint”. Here the weights are positive, so the denominator is positive on the whole feasible cone and only their first program is ever needed, while a negative *numerator* is representable inside it. That is an observation about the instance class, not a device, and it should not be presented as one.

8.2 The computational axis

The project’s own experimental evidence does not support a performance claim. On the frozen test bed the max-flow backends — parametric decomposition and Dinkelbach — solve more instances, and solve them faster, than ratio-closure simplex in every configuration tried; the numbers are in the experimental report and are not reproduced here, because they are being re-measured on a quiet machine and this document must not carry a second, drifting copy of them. The qualitative finding is stable across every campaign run so far and is unlikely to move: the dominant cost is degeneracy, the primal spends the overwhelming majority of its pivots on zero steps, and the dual removes the degeneracy without buying a comparable speedup, trading null steps for small ones.

The consequence for positioning is that the work is, today, a *specialisation* paper and not a *performance* paper, and the three experiments that could change that have not been run:

- E1. **Single ratio oracle.** ratio-closure simplex against a spanning-forest linearization implementation adapted to the first-chunk task, against a static min-cut/root search, and against a general LP solver.
- E2. **Full decomposition path.** Repeated residual oracles against spanning-forest linearization, against parametric max-flow (Gallo et al., 1989), against the parametric Lerchs–Grossmann of Hochbaum (2001), and against a mining-style implementation (Byun and Dimitrakopoulos, 2013) where available.
- E3. **Sequences of related instances.** Vertex deletions, forced inclusions and exclusions, residual prefix removal, small profit and weight perturbations, arc additions and removals — comparing forest basis reuse against pseudoflow warm start, against parametric state reuse, and against cold restart.

E2 was the natural place to look for an advantage, and it has now been measured on the full canonical sequence rather than on a single capacity. The result is negative and the reason is instructive, so both are recorded here.

The task **full-path** was run on twelve instances spanning the generated families, a historical PCKP instance and a MineLib instance, comparing the tuned primal, the dual, Dinkelbach and the parametric decomposition, two replicas each, under a sixty-second limit. The parametric backend is fastest on eleven of the twelve; the dual wins only on the arc-free graph, where there is nothing to pivot on. The margin is not marginal: the median ratio between the tuned primal and the parametric backend is a factor of twelve, and on a layered DAG with $n = 2000$ it is a factor of 859 — sixty seconds against seven hundredths. On the largest instance tried, a sparse DAG with $n = 20,000$, the tuned primal, the dual and Dinkelbach all exhaust the limit while the parametric backend finishes the whole sequence in 3.6 seconds.

The diagnosis matters more than the ranking. The oracle count is exactly what the design predicts — one ratio oracle per closure layer, the same count Dinkelbach pays — and the parametric backend uses about two flow computations per layer. The difference is entirely inside the oracle: each one costs tens to hundreds of thousands of pivots, of which between 98% and 100% are degenerate. On the layered instance the primal spends 768,639 pivots, essentially all of them zero steps, to produce a sequence the parametric backend obtains with a few hundred flow computations.

The warm start was then implemented and measured, and it recovers most of that cost. The extracted block *is* the positive component, that is one tree of the basis forest, so the other trees survive untouched and stay anchored: the next oracle can reuse that forest instead of rebuilding from the empty basis. The condition that was missing from the project’s earlier decision is a single one and is checked rather than proved: the values are $1/w(T)$ on the positive tree T and zero elsewhere, so an active arc leaving T would violate feasibility and T must be closed in the

residual. The best closed surviving tree is chosen, and if none is closed the oracle restarts cold. The check costs $\mathcal{O}(m)$ per oracle.

The effect is on the pivot count, which is where the time was: 251,780 pivots become 3,080 on a disconnected DAG, 89,468 become 6,108 on a sparse one, 48,974 become 6,915 on `minelib_newman1`. The decomposition produced is identical block by block and the objective of a capacity query is unchanged to nine decimals. In wall time the warm primal is 3.1 times faster than the cold one, overtakes Dinkelbach, and wins three of the ten instances every configuration solves — the families whose canonical sequence is longest — against one of twelve before.

This changes the size of the gap, not its direction. The parametric backend still wins the other seven instances and is 3.4 times faster in total, so the positioning of Section 7 stands: this is a specialisation paper, not a performance paper. What has changed is that the axis is no longer hypothetical. The degeneracy is reduced, not removed, and two measurements are still owed: the frozen configuration was tuned on single-capacity solves and never on the full decomposition — a paired ablation on this task puts a sink initial basis 1.21 times ahead of it — and the anti-cycling regime has never been tuned for this task at all.

An end-to-end branch-and-cut comparison comes after that, not before, and must not be used as the first piece of evidence.

8.3 Honest summary

For the forest structure, the basic-solution characterisation, the merge/split pivots, the sign test on branch aggregates, maximum-ratio closure, the canonical decomposition and the incidence-matrix-plus-side-row paradigm, ratio-closure simplex is not new — and the forest itself is not new since 1965, not since 2025. What may be new is the specialisation of the ordinary simplex, primal and dual, to this particular linear program, the closed forms that specialisation produces, and the use of it to drive the entire canonical sequence of blocks rather than the first — and that is a narrow claim which Section 7 states narrowly, and which Q1, Q2 and Q4 have to close before it is made in public — Q3 and Q5 having been closed by the readings recorded above, both of them against the claim rather than for it.

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